Meghan D. Hutchins

Phone: 608.358.8332 | Email: mhutch@umich.edu | Website: http://meghanhutchins.com

Objective

Seeking a PhD in computational neuroscience beginning Fall 2027.

Research Interests

Computational neuroscience and complex systems: how local interactions among units produce emergent behavior at scale. Mechanistic modeling of neural circuit dynamics underlying behavior, learning, and prediction.

Education

2008 - 2010 **University of Michigan** Ann Arbor, MI
M.S. Environmental Informatics
Graduate Certificate: Complex Systems

1997 - 2002 **University of Wisconsin–Eau Claire** Eau Claire, WI
B.S. Computer Science

Research Experience

2008 - 2011 **University of Michigan** Ann Arbor, MI
Agent-Based Modeling Research Specialist

Co-authored 6 peer-reviewed papers and 6 conference proceedings (see Publications)

School of Natural Resources & Environment | Oct 2008 - Dec 2011
Designed and implemented a multi-scale agent-based modeling framework simulating exurban land-use change and carbon storage. Contributed to a large interdisciplinary research team across multiple peer-reviewed publications and conference presentations.

School of Public Health | Sep 2009 - Oct 2011
Developed agent-based models to evaluate the performance of statistical methods for detecting network influence in longitudinal data.

Inter-University Consortium for Political and Social Research | Sep 2010 - Oct 2011
Developed an agent-based model simulating farm-level agricultural decision-making and historical land-use change, validated against household-level historical records.

Master's Thesis | Sep 2008 - Dec 2010
Built ELMST, an agent-based model of how neighbors' influence on yard management affects carbon storage in residential subdivisions. Served as the proof-of-concept for the NSF-funded SLUCE2 project. Found that neighbor-to-neighbor influence lowered mean neighborhood carbon storage and roughly doubled the spread of outcomes versus isolated management, with turf management the dominant driver.

Training in Computational Neuroscience

2026 - **Neuromatch Academy** | Computational Neuroscience - Preparatory Materials
Python, linear algebra, calculus, statistics, dynamical systems, stochastic processes, ML.

2026 **Introduction to Neuroscience |** Dr. Bing Wen Brunton
Quarter-long YouTube course, approximately 30 hours.

Publications

Peer-Reviewed Journal Articles

Sauser Zachrison, K., Iwashyna, T. J., Gebremariam, A., **Hutchins, M**., & Lee, J. M. Can longitudinal generalized estimating equation models distinguish network influence and homophily? An agent-based modeling approach to measurement characteristics. *BMC Medical Research Methodology* (2016), 16, 174.

Currie, W. S., Kiger, S., Nassauer, J. I., **Hutchins, M**., Marshall, L. L., Brown, D. G., Riolo, R. L., Robinson, D. T., & Hart, S. K. Multi-scale heterogeneity in vegetation and soil carbon in exurban residential land of southeastern Michigan, USA. *Ecological Applications* (2016), 26 (5), 1421–1436.

Sylvester, K. M., Brown, D. G., Leonard, S. H., Merchant, E., & **Hutchins, M**. Exploring agent-level calculations of risk and returns in relation to observed land-use changes in the US Great Plains, 1870–1940. *Regional Environmental Change* (2015), 15 (2), 301–315.

Sun, S., Parker, D. C., Huang, Q., Filatova, T., Robinson, D. T., Riolo, R. L., **Hutchins, M.**, & Brown, D. G. Market impacts on land-use change: An agent-based experiment. *Annals of the Association of American Geographers* (2014), 104 (3), 460–484.

Nassauer, J. I., Cooper, D. A., Marshall, L. L., Currie, W. S., **Hutchins, M.**, & Brown, D. G. Parcel size related to household behaviors affecting carbon storage in exurban residential landscapes. *Landscape and Urban Planning* (2014), 129, 55–64.

Robinson, D. T., Sun, S., **Hutchins, M.**, Riolo, R. L., Brown, D. G., Parker, D. C., Filatova, T., Currie, W. S., & Kiger, S. Effects of land markets and land management on ecosystem function: A framework for modelling exurban land-change. *Environmental Modelling & Software* (2013), 45, 129–140.

Conference Proceedings & Abstracts

Currie, W., Brown, D. G., Brunner, A., Fouladbash, L., Hadzick, Z., **Hutchins, M.**, Kiger, S. E., Makino, Y., Nassauer, J. I., Robinson, D. T., Riolo, R. L., & Sun, S. Incorporating ecosystem processes controlling carbon balance into models of coupled human-natural systems. *AGU Fall Meeting Abstracts* (2012), B53F-0752.

Sun, S., Parker, D. C., Brown, D. G., Huang, Q., Filatova, T., Robinson, D. R., **Hutchins, M.**, & Riolo, R. Explicitly representing heterogeneous land developers in agent-based modeling of land use change: A preliminary experiment. *Proceedings of the AAG Annual Meeting* (2012).

Robinson, D. T., Filatova, T., Sun, S., Riolo, R. L., Brown, D. G., Parker, D. C., **Hutchins, M.**, Currie, W. S., & Nassauer, J. I. Integrating land markets, land management, and ecosystem function in a model of land change. *5th International Congress on Environmental Modelling and Software* (2010), Ottawa, Canada. https://scholarsarchive.byu.edu/iemssconference/2010/all/75

Brown, D., Robinson, D., Parker, D., Riolo, R., Currie, W., **Hutchins, M.**, Filatova, T., Kiger, S., Nassauer, J., & Page, S. Framework for modeling effects of land use and land management processes on vegetation productivity and carbon storage in exurban Southeastern Michigan. *GLP Open Science Meeting 2010 Land Systems, Global Change and Sustainability, Book of Abstracts* (2010), 94.

Kiger, S., Currie, W. S., Brown, D. G., Riolo, R., **Hutchins, M.,** & Robinson, D. Landscape carbon and nitrogen trajectories in the decades following conversion to exurban residential development. *95th ESA Annual Meeting* (2010).

Hutchins, M. ELMST: An agent-based model of residential land management and carbon storage. *Poster presentation, AAG Annual Meeting* (2010), Washington, DC.

Thesis

Hutchins, M. Exploring the effects of yard management and neighborhood influence on carbon storage in residential subdivisions. *Master's Thesis, University of Michigan* (2010). https://hdl.handle.net/2027.42/78211

Employment

2025 - 2026 **William & Mary - Office of the University Registrar** Williamsburg, VA
Programmer & Systems Analyst

Wrote Docker containerized Python and SQL programs to extract and process data from the Ellucian Banner database. Generated structured reports. Automated data collection to allow Registrars to run their own reports via Airflow.

2011 - 2025 **Jamf** Minneapolis, MN
Software Engineering Senior Manager & Full-Stack Developer

Community Platform & Team Founder (Jamf Nation) | Oct 2011 - Oct 2014
Architected and coded the Jamf Nation community platform from inception as the sole engineer for two years, scaling it into the world's largest peer-led Apple IT network. Built, hired, and led the dedicated engineering team.

Strategic Leadership (Jamf Now & Jamf Protect) | Oct 2014 - May 2025
Formulated engineering roadmaps and led cross-functional teams across distinct enterprise product lines. Successfully scaled the Jamf Now (Device Management) engineering ecosystem before pivoting to direct the engineering team for *Jamf Protect* (Endpoint Security).

2002 - 2008 **University of Wisconsin Hospital and Clinics** Madison, WI
Full-Stack Web Application Developer

Built and maintained web-based Java applications. Designed and developed applications based on specifications. Wrote and optimized functional SQL.

Community & Outreach

2015 - 2026 Williamsburg Pipes & Drums – Bagpiper, Instructor, Webmaster, Board Member (President 2019–23)
2019 - 2024 Women in Tech Mentor (Jamf)
2018 - 2024 Toastmasters International – Member, VP of Public Relations & Sergeant at Arms
2013 - 2016 Jamf Nation Global Foundation – Founding Board Member
2009 - 2011 Agent-Based Modeling Support Group (University of Michigan) – Founder & Organizer
2009 - 2011 Matthaei Botanical Gardens & Nichols Arboretum - Live-in Caretaker and Educator
2009 - 2010 Environmental Informatics Track Leader (University of Michigan)
2000 - 2001 UW-Eau Claire Student Radio – DJ (curated funk and variety shows)